IMEN574 / IDSC574 · PROGRAMMING FOR AX

Review Session

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Spring 2027

Outline: Review Session

01 **AI and Data**

Check the request and the input

02 **Supervised Learning**

Connect objectives to decisions

03 **Unsupervised Learning and Interpretation**

Read structure without overclaiming

01 · AI and Data

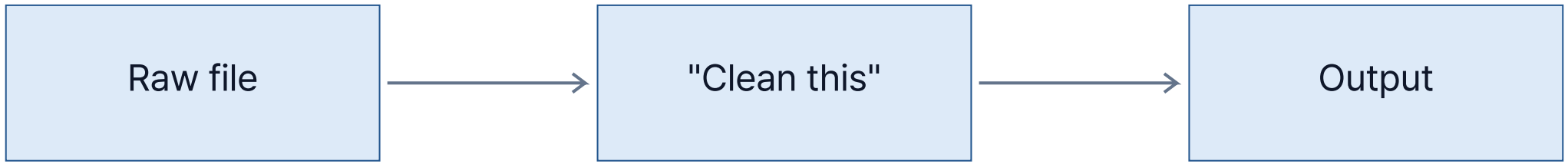
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Question: A Checkable Request

What information makes this task testable?



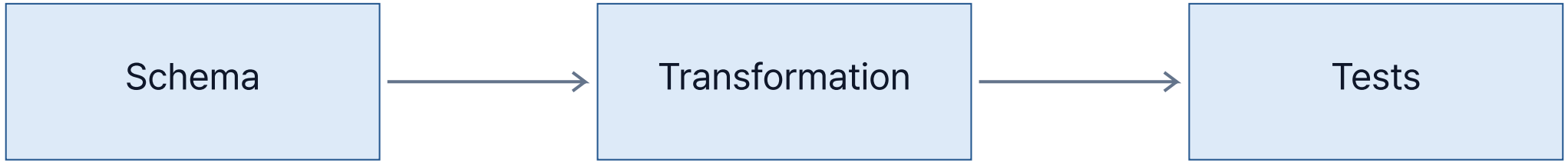
missing + duplicate IDs

no criteria

unknown validity

Answer: A Checkable Request

Specify the input, output, and acceptance tests



row = one review

preserve IDs and order

missing + repeated ID

Question: A Decision Boundary

Which input distinguishes > from ≥?

Score	Label
0.2	0
0.5	?
0.8	1

Answer: A Decision Boundary

The equality case tests the specification

Condition	Result
$0.5 > 0.5$	False
$0.5 \geq 0.5$	True

Demo: Mutable Aliases

What changes when b appends a value?

Predict the result, then run; change y = x to y = x[:] and run again

Python

```
x = [1, 2]
y = x
y.append(3)
print(x)
print(y)
```

Run Python Reset

```
[1, 2, 3]
[1, 2, 3]
```

Answer: Mutable Aliases

Both names refer to the same list

Name	Value
x	[1, 2, 3]
y	[1, 2, 3]

Question: Broadcasting Shapes

Can (3, 1) and (1, 2) be added?

x

1
2
3

Row vector (1,2)

[10, 20]

Answer: Broadcasting Shapes

Trailing dimensions expand to a (3, 2) result

+10	+20
11	21
12	22
13	23

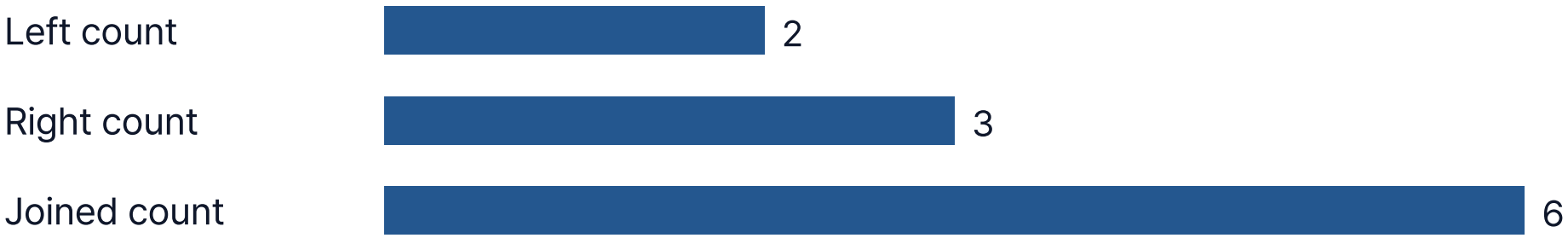
Question: Repeated Join Keys

How many joined rows does this key produce?

Key	Side
A	left 1
A	left 2
A	right 1
A	right 2
A	right 3

Answer: Repeated Join Keys

Two left rows meet three right rows



Question: Imputation Leakage

Which mean fills a missing training value?

Split	Observed x
train	2
train	4
test	100

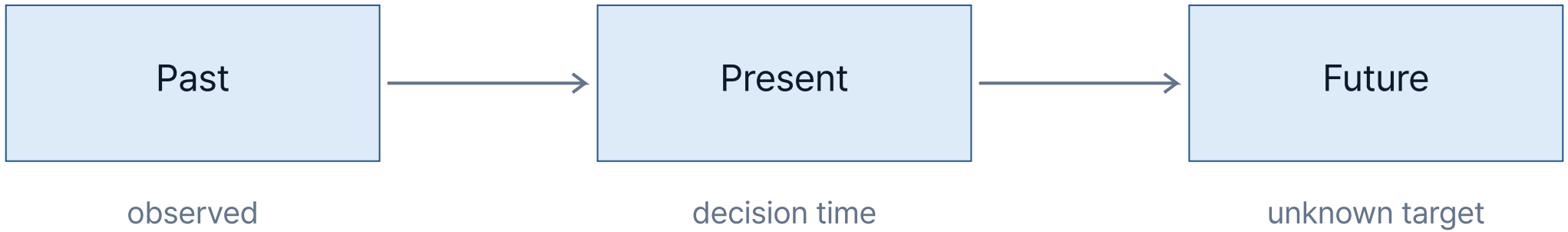
Answer: Imputation Leakage

Estimate the imputer on training rows only



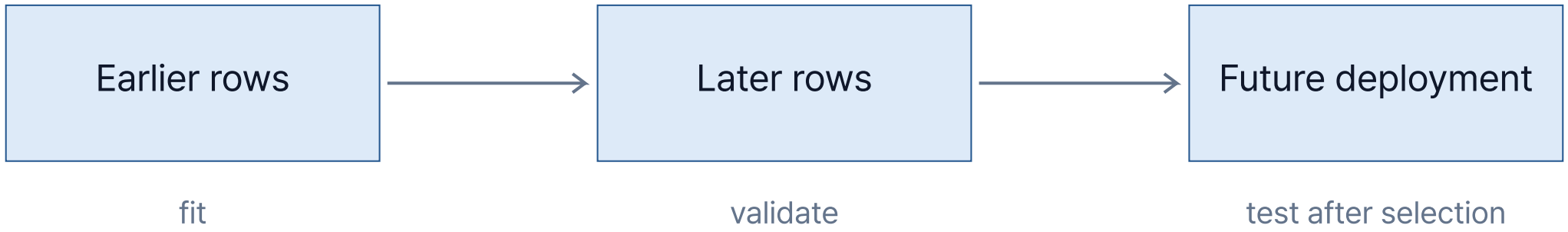
Question: Time-Aware Evaluation

Which split imitates a forecast?



Answer: Time-Aware Evaluation

Training precedes the held-out future



02 · Supervised Learning

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Read structure without overclaiming

Question: A Regression Baseline

What is the mean predictor for y = [2, 4, 6]?

y
2
4
6

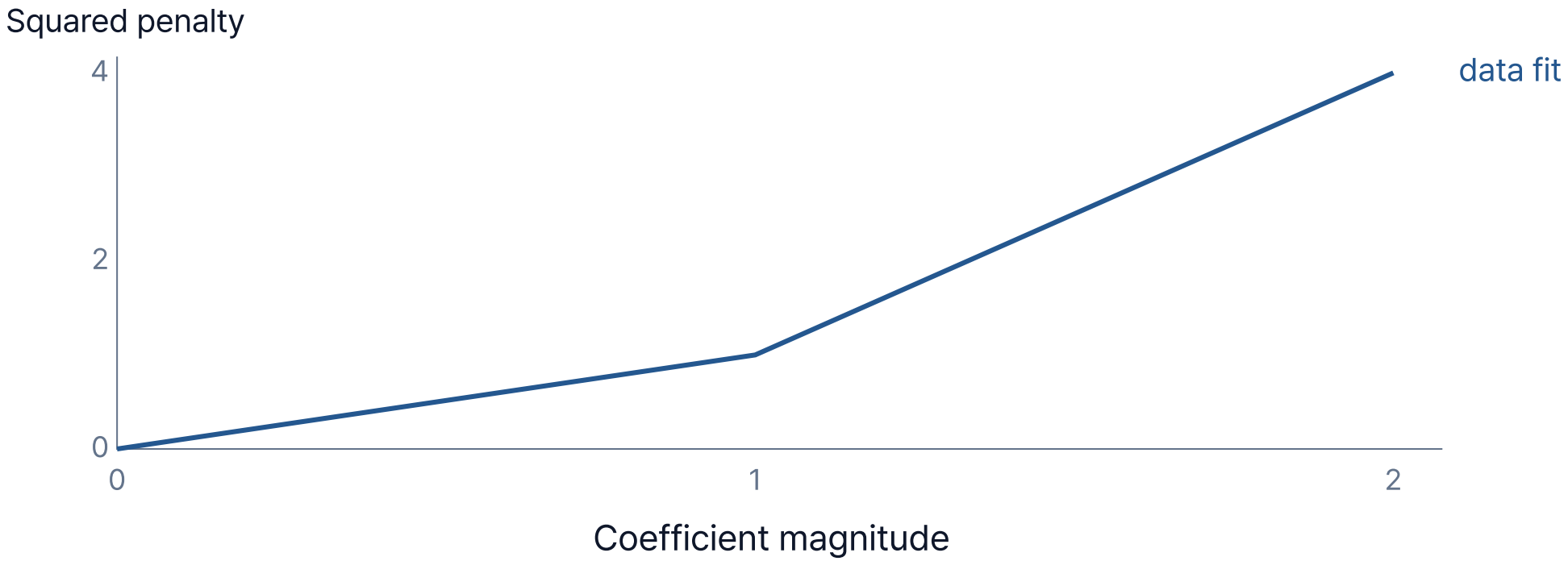
Answer: A Regression Baseline

The constant least-squares prediction is 4

y	Mean	Residual
2	4	-2
4	4	0
6	4	2

Question: A Ridge Penalty

What happens when the penalty λ increases?



Answer: A Ridge Penalty

The no-intercept estimate shrinks toward zero

Setting	Calculation
$X = [1, 2]$, $y = [2, 4]$	$\Sigma xy = 10$; $\Sigma x^2 = 5$
$\lambda = 0$	$w = 10/5 = 2$
$\lambda = 5$	$w = 10/10 = 1$

Question: Precision and Recall

Compute precision and recall from this matrix

	Pred 1	Pred 0
True 1	3	1
True 0	1	3

Answer: Precision and Recall

Precision = 3/4 and recall = 3/4



Question: Imbalanced Accuracy

Is 99% accuracy enough evidence?



Answer: Imbalanced Accuracy

Predicting every row negative has zero positive recall

Metric	All-Negative Predictor
Accuracy	990/1000 = 0.99
Positive recall	0/10 = 0

Question: Boosting Residuals

What target follows the initial constant prediction?

y	Initial prediction
2	4
4	4
6	4

Answer: Boosting Residuals

The next stage fits residuals [−2, 0, 2]

y	f_0	$y - f_0$
2	4	-2
4	4	0
6	4	2

03 · Unsupervised Learning and Interpretation

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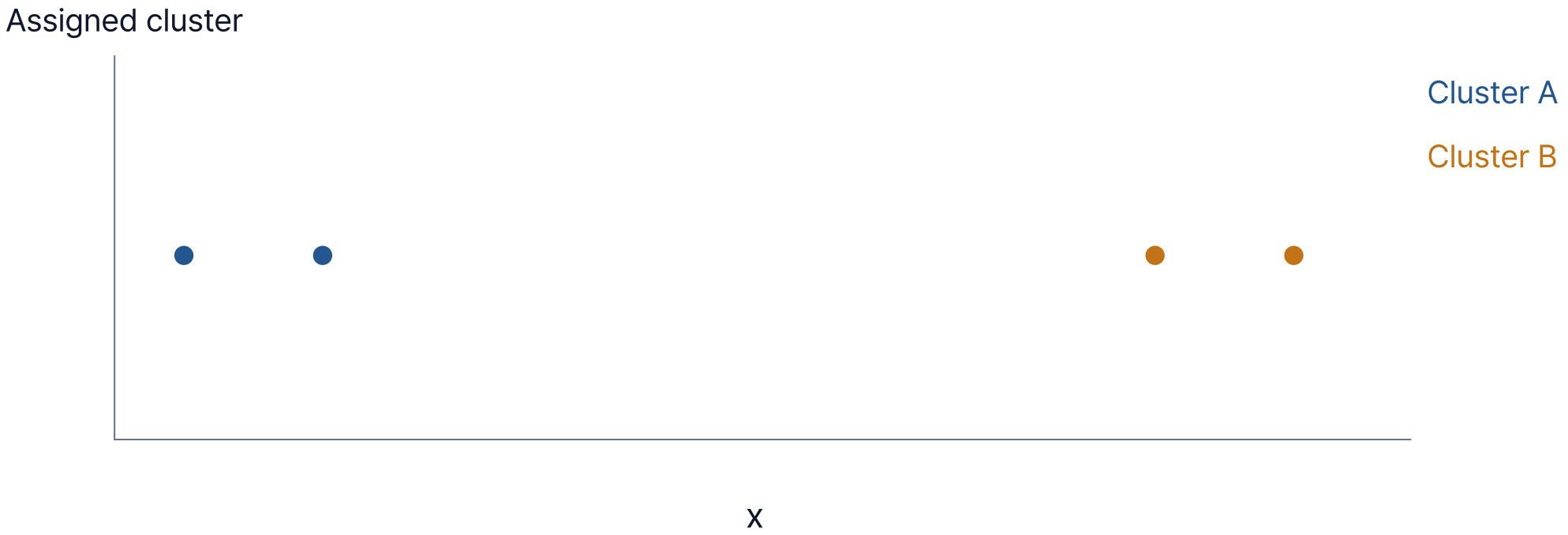
Connect objectives to decisions

03 **Unsupervised Learning and Interpretation**

Read structure without overclaiming

Question: K-Means Centers

Update the centers after these assignments



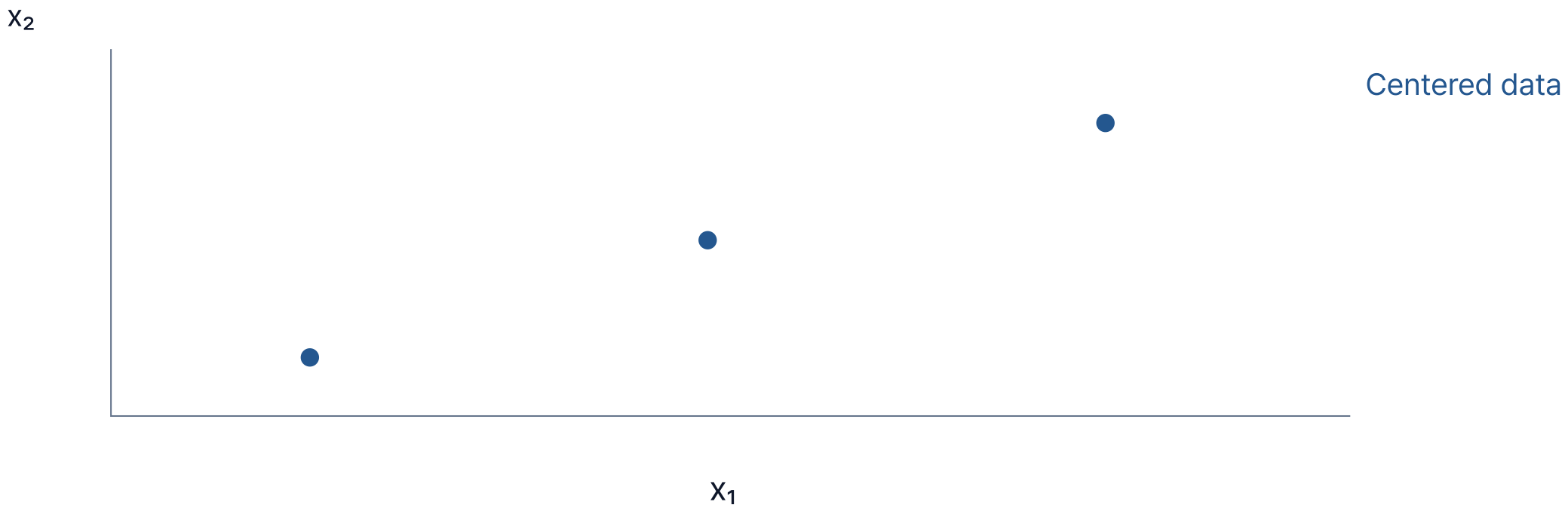
Answer: K-Means Centers

Each center becomes its assigned mean

Cluster	Updated center
A	(1+2)/2 = 1.5
B	(8+9)/2 = 8.5

Question: A Principal Direction

Which direction preserves this variation?



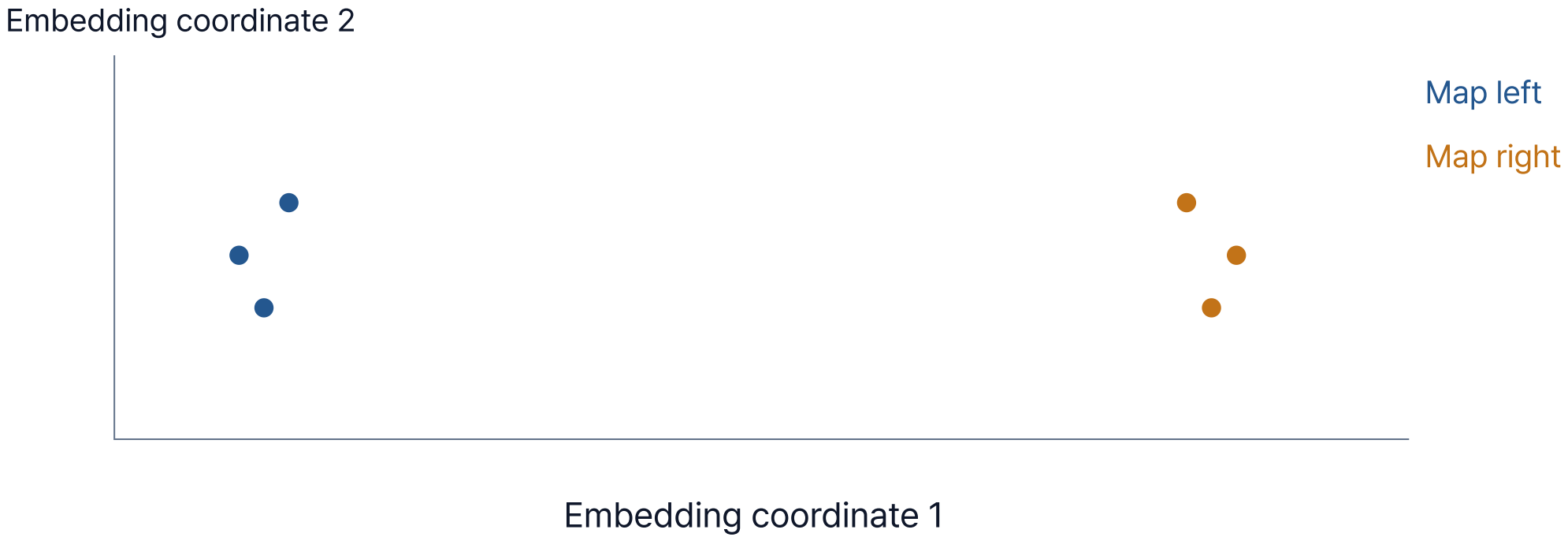
Answer: A Principal Direction

Project along (1, 1) / √2

Point	One-dimensional score
(-1,-1)	−√2
(0,0)	0
(1,1)	√2

Question: A Two-Dimensional Map

Does an empty gap establish separate populations?



Answer: A Two-Dimensional Map

A visualization needs stability and original-space checks



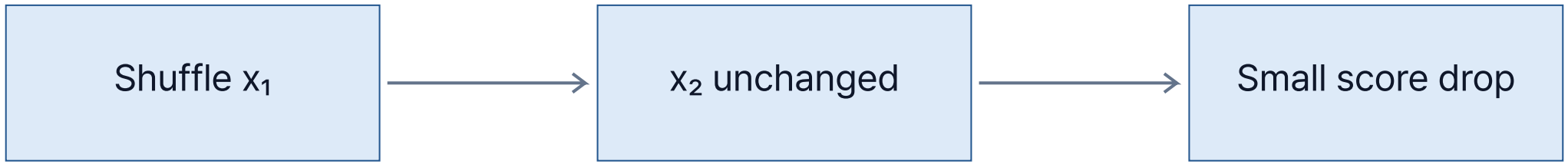
Question: Correlated Features

What does permutation measure when another feature duplicates the signal?

x_1	x_2 copy	y
1	1	0
2	2	0
8	8	1
9	9	1

Answer: Correlated Features

The model may retain the signal through the unshuffled copy



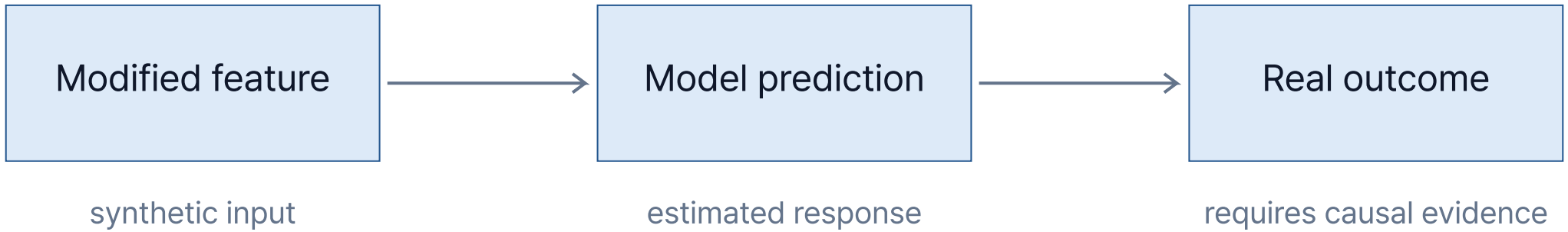
break one column

same signal remains

not proof of irrelevance

Question: An Explanation and a Cause

Does changing a feature in a model identify a real intervention effect?



Answer: An Explanation and a Cause

Predictive explanations do not establish intervention effects

Question	Target
Model question	What prediction changes?
Causal question	What real outcome changes?

Takeaway: Review Session

01 Specify what counts as correct

02 Keep evaluation outside fitting

03 Interpret evidence within its scope